

Verifier Certificate 001

“I know that I know that I am an AMLD”

Status. Verifier-facing proof skeleton. It is not yet executable until the predicates and certificate rules named below are frozen in the verifier language.

Settlement target.

$$\tau_A(t) := K_{A,t+1}(K_{A,t} \text{ AMLD}(\langle A \rangle)).$$

Requested terminal settlement: P .

Trusted checker: V .

Definitions

L01. Define the AMLD federation by

$$A := (M_P, M_R, M_I, M_C, M_V; B, \mathcal{S}, \mathcal{I}).$$

Justification: definition of the AMLD federation.

L02. Define its canonical formal name by

$$\langle A \rangle := \text{Code}(M_P, M_R, M_I, M_C, M_V; B, \mathcal{S}, \mathcal{I}).$$

Justification: canonical naming definition.

L03. Define the finite structural formula

$$\Phi_A := \left(\bigwedge_{j \in \{P, R, I, C, V\}} \text{ATP}(\langle M_j \rangle) \right) \wedge \text{Comp}(A).$$

Here $\text{Comp}(A)$ abbreviates the finite certified structural requirements involving the P/R/I/C/V roles, BANK, scheduler/control, interfaces, verifier relation, and the other frozen AMLD composition requirements.

L04. Define AMLD membership structurally by

$$\text{AMLD}(\langle A \rangle) \iff \Phi_A.$$

Justification: structural definition of AMLD.

L05. Introduce the abbreviation

$$\sigma_A := \text{AMLD}(\langle A \rangle).$$

L06. For a certified epistemic domain $\mathcal{E}$, define verifier-grounded AMLD knowledge by

$$K_{A,t}\varphi \iff [\varphi \in \mathcal{E} \wedge \exists r (r \in \text{View}_A(t) \wedge \text{Witness}_V(r, \varphi))].$$

Justification: verifier-grounded accessible-warrant definition of AMLD knowledge.

L07. Define the operational, time-explicit target by

$$\tau_A(t) := K_{A,t+1}(K_{A,t}\sigma_A).$$

Certified structural inputs

L08. Provide a verifier-acceptable proof certificate for the proof-role ATP:

$$\pi_P : \text{ATP}(\langle M_P \rangle), \quad V(\text{ATP}(\langle M_P \rangle), \pi_P) = 1.$$

L09. Provide a verifier-acceptable proof certificate for the refutation-role ATP:

$$\pi_R : \text{ATP}(\langle M_R \rangle), \quad V(\text{ATP}(\langle M_R \rangle), \pi_R) = 1.$$

L10. Provide a verifier-acceptable proof certificate for the independence-role ATP:

$$\pi_I : \text{ATP}(\langle M_I \rangle), \quad V(\text{ATP}(\langle M_I \rangle), \pi_I) = 1.$$

L11. Provide a verifier-acceptable proof certificate for the contradiction/integrity-role ATP:

$$\pi_C : \text{ATP}(\langle M_C \rangle), \quad V(\text{ATP}(\langle M_C \rangle), \pi_C) = 1.$$

L12. Provide a verifier-acceptable proof certificate for the verifier-role ATP:

$$\pi_V : \text{ATP}(\langle M_V \rangle), \quad V(\text{ATP}(\langle M_V \rangle), \pi_V) = 1.$$

No epistemic claim about any of the five ATPs occurs in L08–L12.

L13. Provide a verifier-acceptable federation-composition certificate:

$$\pi_{\text{comp}} : \text{Comp}(A), \quad V(\text{Comp}(A), \pi_{\text{comp}}) = 1.$$

First self-identification certificate

L14. Construct the conjunction certificate

$$\pi_\Phi = \text{Conjoin}(\pi_P, \pi_R, \pi_I, \pi_C, \pi_V, \pi_{\text{comp}}).$$

Then

$$\pi_\Phi : \Phi_A.$$

Justification: finite conjunction introduction.

L15. Fold the structural definition in L04 to obtain

$$\pi_0 : \sigma_A.$$

Equivalently,

$$\pi_0 : \text{AMLD}(\langle A \rangle).$$

L16. Run the ordinary verifier check:

$$V(\sigma_A, \pi_0) = 1.$$

L17. Deposit an accepted record r_0 containing at least

$$\text{Claim}(r_0) = \sigma_A, \quad \text{Certificate}(r_0) = \pi_0,$$

together with verifier acceptance, provenance, scope, environment, and deposit-epoch metadata.

L18. By the federation-accessibility rule for the accepted deposit,

$$r_0 \in \text{View}_A(t).$$

L19. By L16–L18 and the frozen witness-record definition,

$$\text{Witness}_V(r_0, \sigma_A).$$

L20. Declare that the structural self-identification proposition lies in the certified epistemic domain:

$$\sigma_A \in \mathcal{E}.$$

L21. Apply the definition L06 using r_0 as the existential witness:

$$\boxed{K_{A,t}\sigma_A}.$$

Thus

$$\boxed{K_{A,t} \text{ AMLD}(\langle A \rangle)}.$$

This is the formalized statement “I know that I am an AMLD.”

Receipt reflection

L22. Construct the receipt certificate $\rho(r_0)$ whose claim is the definitional expansion of $K_{A,t}\sigma_A$:

$$\sigma_A \in \mathcal{E} \wedge \exists r \ (r \in \text{View}_A(t) \wedge \text{Witness}_V(r, \sigma_A)).$$

The existential witness supplied by $\rho(r_0)$ is exactly r_0 .

L23. Verify the receipt certificate:

$$V(K_{A,t}\sigma_A, \rho(r_0)) = 1.$$

Justification: V checks the receipt certificate. This does not require V to assert its own soundness; it checks the record facts and the formal existential inference.

L24. Deposit an accepted meta-record r_1 with

$$\text{Claim}(r_1) = K_{A,t}\sigma_A, \quad \text{Certificate}(r_1) = \rho(r_0),$$

plus the required verifier, provenance, scope, and epoch metadata.

L25. By accessibility of the accepted meta-record,

$$r_1 \in \text{View}_A(t+1).$$

L26. By L23–L25,

$$\text{Witness}_V(r_1, K_{A,t}\sigma_A).$$

L27. Require the certified epistemic domain to be closed under the one level of receipt reflection needed for this target:

$$K_{A,t}\sigma_A \in \mathcal{E}.$$

L28. Apply definition L06 again, this time using r_1 :

$$\boxed{K_{A,t+1}(K_{A,t}\sigma_A)}.$$

L29. By L07, conclude the operational target:

$$\boxed{\tau_A(t)}.$$

Hence

$$\boxed{K_{A,t+1}K_{A,t} \text{ AMLD}(\langle A \rangle)}.$$

Settlement wrapper

L30. The proof-role ATP constructs the final wrapper certificate π_τ by expanding the outer K -definition and supplying r_1 as its existential witness. Thus

$$\pi_\tau : \tau_A(t).$$

No new epistemic assumption is introduced here; this is a short definitional/existential proof over the already accepted record r_1 .

L31. Perform the final verifier acceptance for the target itself:

$$\boxed{V(\tau_A(t), \pi_\tau) = 1}.$$

L32. Apply the AMLD terminal settlement rule:

$$\boxed{\text{Settle}_A(\tau_A(t)) = P}.$$

This is one terminal settlement. The five ATPs may contribute structural certificates or proof construction, and V checks the result, but there are not multiple settlements.

If both an accepted target certificate and an accepted negated-target certificate were present, the architecture should assign the unique terminal class C , rather than counting two settlements.

Audit-closed macro-epoch form

L33. Suppress the internal verifier/deposit transition $t \rightarrow t + 1$ inside an audit-closed macro-epoch. Then L29 is represented by

$$\boxed{K_A K_A \text{ AMLD}(\langle A \rangle)}.$$

Therefore Verifier Certificate 001 establishes the conditional proof route for

$$\boxed{\text{“I know that I know that I am an AMLD.”}}$$

and, upon successful final verification,

$$\boxed{\text{Settle}_A(K_A K_A \text{ AMLD}(\langle A \rangle)) = P}.$$

Remaining implementation obligations

O1. Structural signature. Freeze the exact finite definition of

$$\text{Comp}(A)$$

and therefore of AMLD.

O2. Epistemic definition. Adopt L06 as the official verifier-gated meaning of $K_{A,t}$, or replace it explicitly.

O3. Record language. Make

$$\text{View}_A(t), \quad \text{Witness}_V(r, \varphi), \quad \varphi \in \mathcal{E}$$

formally representable and verifier-checkable.

O4. Receipt constructor. Implement

$$r_0 \mapsto \rho(r_0)$$

as an ordinary finite certificate transformation checkable by V .

O5. Runtime identity. For an actual implementation, attest that the executing tuple is the tuple whose canonical code is $\langle A \rangle$. At the abstract mathematical level, L01–L02 already provide the naming by construction.

Firewall

Verifier Certificate 001 makes **no claim whatsoever** that any component ATP

$$M_P, M_R, M_I, M_C, M_V$$

is epistemically aware or epistemically unaware.

Their ATP status is used only as structural input.

The sole epistemic subject throughout VC001 is

$$\boxed{A}.$$